VinoIQ: Predicting Wine Quality Using Machine Learning

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1. Problem Description

1.1 Scope

Wine quality is traditionally evaluated through sensory assessments involving characteristics such as taste, aroma, and balance. Although these evaluations provide valuable information, they are subjective and depend on human judgment. This project investigates whether machine learning can estimate wine quality using measurable physicochemical properties.

The objective is to develop VinoIQ, an interactive web application that predicts red wine quality based on chemical measurements. The name combines *vino*, meaning wine, with *IQ*, representing intelligence. The application uses the UCI Wine Quality dataset, containing 1,599 Portuguese red wine samples with 11 physicochemical measurements and recorded sensory quality scores.

VinoIQ is intended primarily as an educational and exploratory tool for students and others interested in practical machine learning applications. Users can enter chemical measurements, receive a predicted quality score, and explore how the model performs. The application is not intended to replace professional wine tasting or provide an objective measure of taste.

Machine learning is suitable for this problem because relationships between chemical properties and sensory quality may be complex and nonlinear. The project follows the main stages of the machine learning lifecycle, including data preparation, model training, validation, evaluation, and deployment.

The project's success is measured by whether the selected model performs better than a simple baseline, produces interpretable predictions, and is accessible through a functioning web application. As an educational prototype, the primary objective is to demonstrate a reproducible machine learning workflow rather than achieve a commercial outcome.

1.2 Metrics

Three regression metrics are used to evaluate model performance: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

MAE measures the average absolute difference between predicted and recorded wine quality scores. It is the primary metric for model selection because it is directly interpretable on the quality scale. For example, an MAE of 0.5 means that predictions differ from recorded scores by half a quality point on average.

RMSE also measures prediction error but assigns greater weight to larger mistakes. It therefore complements MAE by highlighting the impact of substantial prediction errors.

R^2 measures how well a model explains variation in the recorded quality scores compared with predicting the mean. A value of 1 represents perfect predictions, while a value of 0 indicates performance equivalent to predicting the mean on the evaluated dataset.

A Dummy Regressor that predicts the average training quality serves as the baseline. Candidate models are compared using validation MAE, and the selected

model is subsequently evaluated on an independent test set to assess its performance on unseen data.

The web application is evaluated through functional testing, including whether users can enter measurements, generate predictions, load example wines, and view model results without errors.

2. Data

2.1 Dataset

The project uses the red wine subset of the UCI Wine Quality dataset, described by Cortez et al. (2009). It contains 1,599 Portuguese red wine samples, each with 11 physicochemical measurements and a recorded sensory quality score.

The 11 input features are fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol. All features are numerical, allowing them to be used directly as inputs to the regression models.

The target variable, *quality*, represents a sensory assessment on a scale from 0 to 10. However, the observed quality scores in this dataset range from 3 to 8. Because the target is numerical, the problem is treated as a supervised regression task.

2.2 Data Preparation

The dataset was loaded using Pandas and checked for missing values. No missing values were found, so imputation was unnecessary. The target variable, *quality*, was separated from the 11 numerical input features.

The data was randomly divided into three subsets: 959 samples (60%) for training, 320 samples (20%) for validation, and 320 samples (20%) for testing. The training set was used to fit the candidate models, the validation set to select the best-performing model, and the independent test set to evaluate its performance on unseen observations.

A fixed random seed (*random_state=42*) was used to support reproducibility. The original features were used without additional feature engineering or scaling. Scaling is generally unnecessary for Random Forest, but its absence may have affected the performance of Ridge Regression.

After model selection, the selected model was retrained using the combined training and validation sets before being evaluated on the test set.

2.3 Data Distribution and Limitations

The distribution of recorded quality scores is uneven, with most wines receiving scores of 5 or 6. Wines with very low or high quality scores are considerably less common, as illustrated in Figure 1. This imbalance may make extreme scores more difficult to predict because the model has fewer examples of these wines.

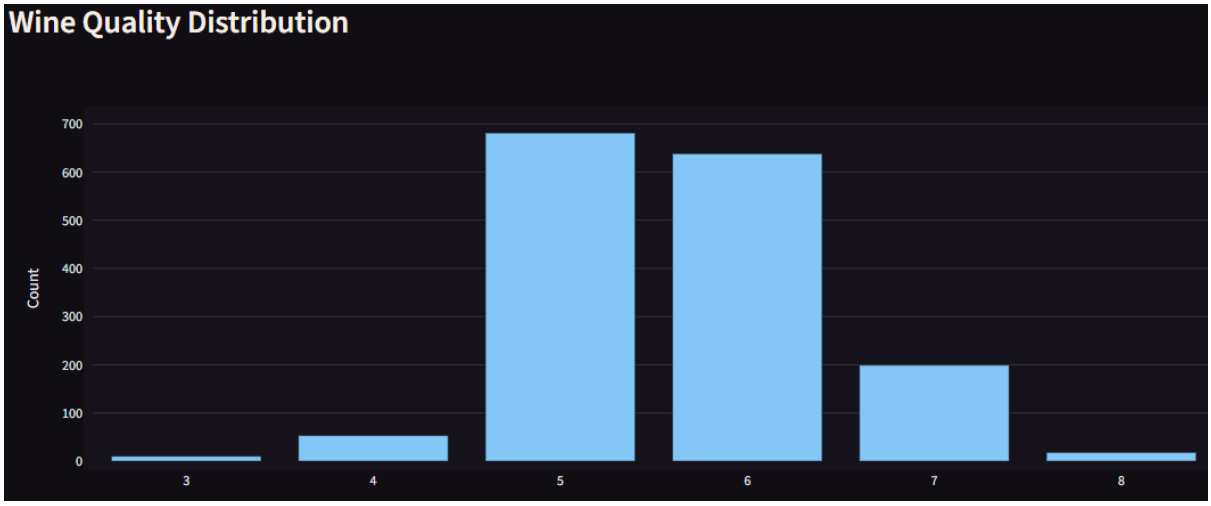


Figure 1. Distribution of recorded wine quality scores in the red wine dataset (n = 1599).

The dataset also has limitations regarding its scope. It contains only Portuguese red wines and does not include information about grape variety, production methods, or individual preferences. Furthermore, the target scores are based on sensory evaluations, which involve subjective human judgment.

Although the dataset does not contain personal information, its limited representativeness and subjective labels affect how the results should be interpreted. The model estimates recorded quality scores rather than determining an objective measure of wine quality.

3. Modeling

3.1 Model Selection and Training

Three regression models were implemented using Scikit-learn to compare approaches of increasing complexity.

A **Dummy Regressor** was used as a baseline, predicting the mean quality score of the training data regardless of the input measurements. This established a reference for evaluating whether more advanced models learned useful patterns.

Ridge Regression was selected as a regularized linear model, providing a comparison based on approximately linear relationships between chemical measurements and wine quality. The regularization parameter was set to *alpha = 10*.

A **Random Forest Regressor** was used to capture potentially nonlinear relationships and interactions between features. It was configured with 250 decision trees, a minimum of two samples per leaf, and a fixed random seed of 42.

All three models were trained on the same training set and evaluated on the validation set using MAE, RMSE, and R^2. The model with the lowest validation MAE was selected for final training. No systematic hyperparameter optimization was performed.

3.2 Model Evaluation

The models were compared using the validation set, with MAE as the primary selection metric. As shown in Table 1, Random Forest achieved the lowest validation MAE (0.439), outperforming Ridge Regression (0.505) and the Dummy Regressor baseline (0.686).

Model	Validation MAE	Validation R^2
Dummy Regressor	0.686	-0.014
Ridge Regression	0.505	0.286
Random Forest	0.439	0.411

Table 1. Validation performance of the three regression models. Lower MAE indicates better predictive accuracy.

Based on these results, Random Forest was selected and retrained using the combined training and validation data. The final model was then evaluated on the independent test set containing 320 wine samples.

The final model achieved an MAE of 0.431, RMSE of 0.558, and R^2 of 0.524. This means predictions differed from recorded quality scores by approximately 0.43 points on average, while the model explained approximately 52.4% of the variation in the test-set quality scores.

These results demonstrate that the model learned useful relationships between physicochemical measurements and recorded wine quality. However, the remaining

prediction errors indicate that chemical measurements alone do not fully explain sensory quality assessments.

3.3 Prediction Errors and Model Limitations

To investigate prediction errors, the model's predictions were compared with recorded quality scores in the independent test set.

Predicted scores ranged from approximately 4.47 to 7.21, with an average of 5.67. This range was narrower than the recorded quality scores, suggesting that the model tended to predict values closer to the middle of the quality distribution.

The model particularly struggled with high-quality wines. Among the 47 test samples with recorded quality scores of 7 or higher, the average recorded score was 7.11, while the average prediction was 6.44. The MAE for these wines was approximately 0.68, compared with 0.43 across the entire test set.

This tendency to underestimate high-quality wines may partly result from the uneven distribution of training samples and the averaging of predictions across multiple decision trees.

Figure 2 illustrates the relationship between recorded and predicted scores. Predictions closer to the diagonal line indicate smaller errors.

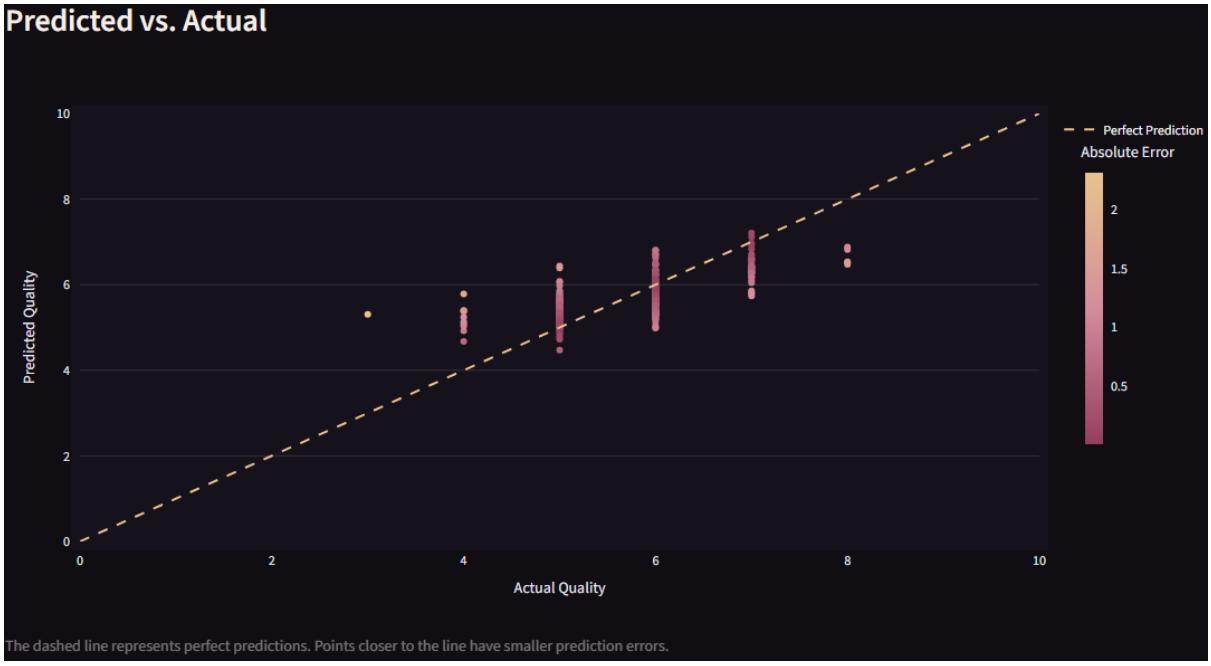


Figure 2. Recorded versus predicted wine quality scores in the independent test set. The dashed line represents perfect predictions.

Overall, the model performed reasonably well for wines with common quality scores but was less reliable for unusually high- or low-quality wines.

3.4 Feature Importance

The Random Forest model provides feature importance scores that estimate how much each input variable contributes to reducing prediction error within the decision trees.

These scores help identify which physicochemical measurements the model relies on most. As shown in Figure 3, alcohol has the highest feature importance, followed by sulphates and volatile acidity. This suggests that these measurements were particularly useful for predicting the recorded wine quality scores in the dataset.

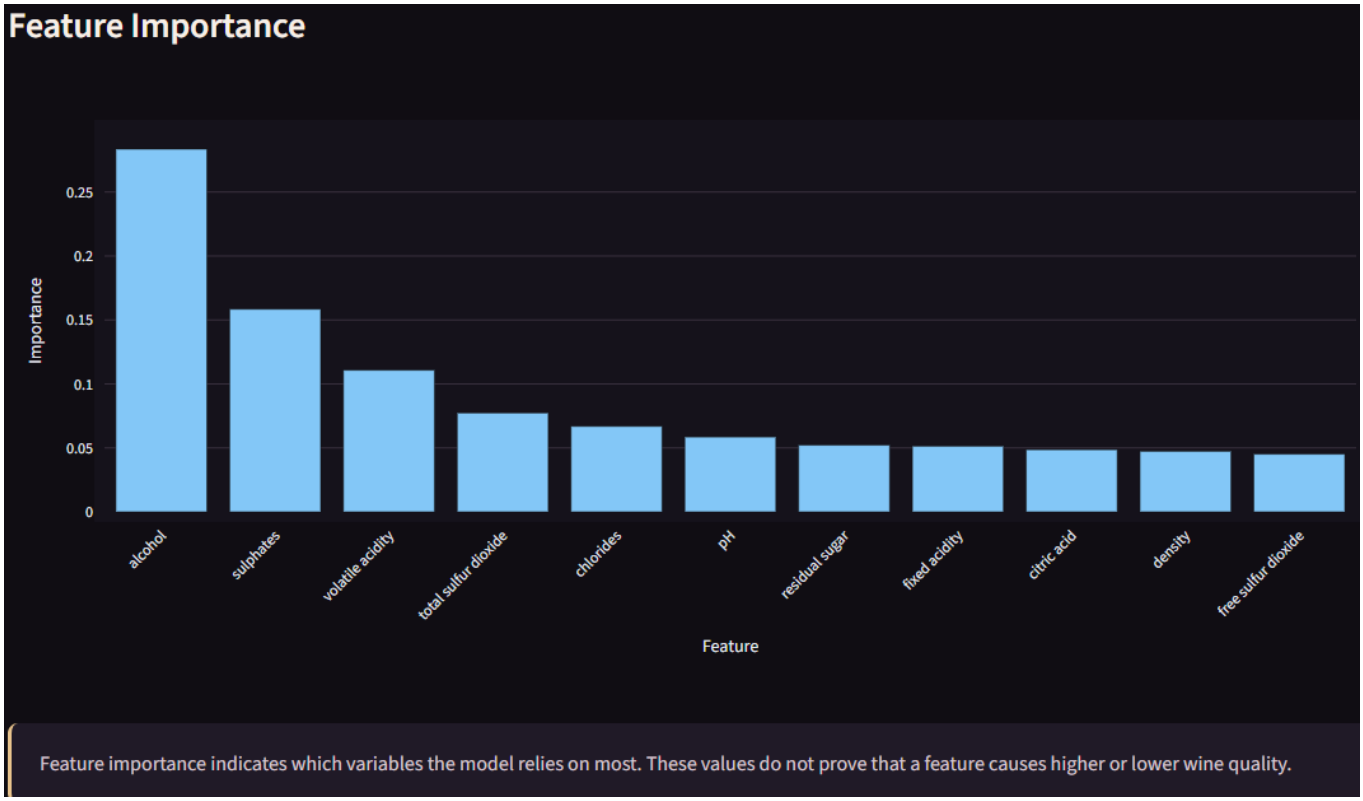


Figure 3. Feature importance scores from the trained Random Forest model.

However, feature importance should not be interpreted as evidence of causation. A high importance score indicates that a variable was useful for prediction, not that changing the chemical property would necessarily improve wine quality.

Future improvements could include systematic hyperparameter tuning, feature scaling for Ridge Regression, and additional training data covering a wider range of wine quality scores.

4. Deployment

4.1 Web Application

VinoIQ was developed as an interactive web application using Streamlit, allowing users to access the trained machine learning model through a web browser. The final Random Forest model was saved using Joblib and loaded by the application to generate predictions without retraining the model for each request.

The application consists of four pages: Dashboard, Wine Analyzer, Model Insights, and About. The Dashboard introduces the project and summarizes key information about the dataset and model performance. Wine Analyzer provides 11 sliders corresponding to the physicochemical input features and allows users to generate predictions or load example wines from the dataset. The Wine Analyzer interface and an example prediction are shown in Figure 4.

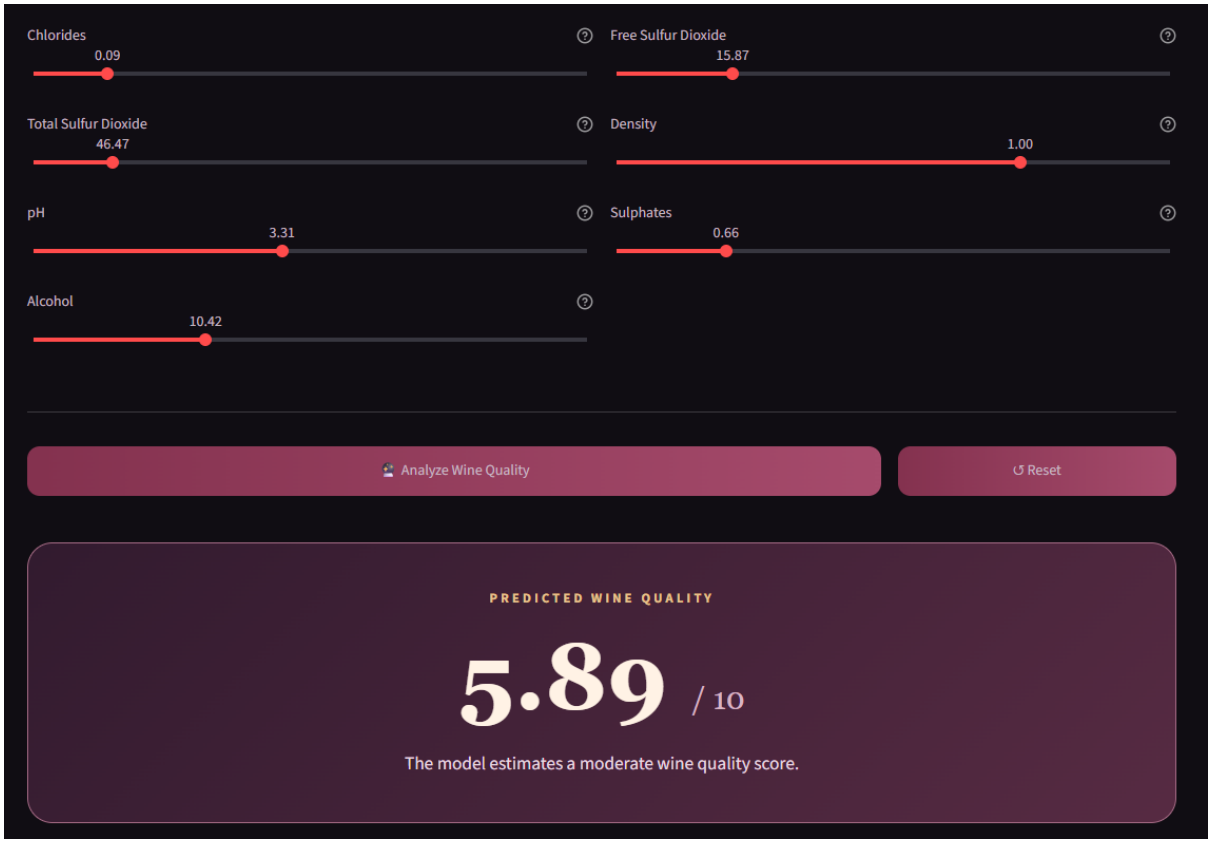


Figure 4. Wine Analyzer interface showing user-adjustable chemical measurements and a predicted wine quality score.

Model Insights presents validation and test results through interactive Plotly visualizations, including model comparisons, prediction errors, and feature importance. The About page explains the project's methodology, data source, and limitations.

The application uses Streamlit for the interface, Scikit-learn for machine learning, Joblib for model storage, Pandas for data handling, and Plotly for data visualization.

4.2 Prediction Workflow

When a user adjusts the sliders in Wine Analyzer, the application collects 11 numerical values representing the chemical properties of a wine. These values are organized into a Pandas DataFrame using the same feature names and order as during model training.

When the user selects Analyze Wine Quality, the saved Random Forest model processes the measurements and returns a numerical prediction. The application displays the estimated quality score, a visual quality scale, and a short interpretation of the result.

Users can also load example wines with recorded quality scores of 3, 5, or 7. If an example is analyzed without changing its original measurements, the application displays the recorded quality alongside the prediction and calculates the absolute error. This provides a practical illustration of how individual predictions may differ from recorded scores.

The prediction workflow connects the trained model to an accessible interface, allowing users to experiment with different wine profiles without programming knowledge.

4.3 Deployment and Maintenance

VinoIQ was deployed using Streamlit Community Cloud and is publicly accessible through a web browser.

Live application: https://vinoiq-dat158.streamlit.app/

GitHub repository: https://github.com/678109/VinoIQ

The application is deployed from the public GitHub repository, which contains the source code, dataset, trained model, evaluation results, and Python dependencies. The repository also includes documentation explaining how to install the dependencies, retrain the model, and run the application locally.

After deployment, the application was tested through its public URL to verify that users could navigate between pages, enter chemical measurements, generate predictions, load example wines, and view model performance visualizations.

The application is intended as an educational prototype rather than a production system. Future maintenance would primarily involve keeping dependencies compatible, verifying that the deployed application remains functional, and retraining the model if suitable new data becomes available. Automated monitoring and retraining were not implemented.

5. References

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6. Appendix A: Declaration of AI Usage

Declaration of AI Usage

Har det i arbeidet med denne rapporten blitt anvendt KI-baserte hjelpemidler?

☒ Ja

☐ Nei

Hvis ja, spesifiser type verktøy og bruksområde nedenfor:
ChatGPT, brukt til å utfordre ideer, og forklare temaer fra pensum

Tekst

☐ Stavekontroll: Er deler av teksten kontrollert av KI-verktøy? (f.eks. Grammarly, Grammar Bot, Laguag Tool, Copilot)

☒ Skriveassistanse: Er deler av strukturen, ideene eller fremgangsmåtene basert på forslag fra KI-verktøy? (f.eks. ChatGPT)

☐ Tekstgenerering: Er deler av teksten generert av KI-verktøy? (f.eks. ChatGPT)

Hvis ja, spesifiser hvilke verktøy og hva dere har brukt de til her:

ChatGPT – Forslag til endringer

Prosess

☐ Idégenerering: Har KI-verktøy blitt brukt til idégenerering i arbeidet med prosjektet?

☒ Sparringspartner: Har KI-verktøy blitt brukt som sparringspartner under prosjektet? (F.eks. for å forstå konsepter fra pensum og forelesninger eller for å få tilbakemeldinger på idéer.)

☐ Informasjonssøk: Har KI-verktøy blitt brukt til informasjonssøk under prosjektet?

Hvis ja, spesifiser hvilke verktøy og hva dere har brukt de til her:

ChatGPT – Forklare temaer enklere, for å hjelpe å forstå pensum

Bilder og figurer

☐ Bildegenerering: Har ett eller flere av bildene eller figurene i rapporten blitt generert av KI-verktøy? (f.eks. Midjourney, Jasper, Dall-E)

Hvis ja, spesifiser hvilke verktøy og hva dere har brukt de til her:

Annet

☐ Andre KI-verktøy

Spesifiser hvilke verktøy og hva dere har brukt de til her: